

Work Division: Pulsar Candidate Classification

UE24CS352A Machine Learning • Mini-Project • Team of 2

Problem: Application of machine learning methods to identify and categorize radio pulsar signal candidates (HTRU2 dataset: 17,898 candidates, 8 features, 1,639 pulsars). The project has two halves: a supervised pipeline (GDA baseline, Random Forest) and an unsupervised pipeline (PCA, K-means, Self-Organizing Map). Each teammate owns one half.

Shared work (Day 1, together)

- Create the private GitHub repo, share it with faculty and TAs, agree on folder structure and commit habits.
- Download HTRU2 and run exploratory analysis: class imbalance, feature distributions, correlations.
- Agree on a common train/test split so results are comparable.

Teammate 1: Supervised classification (pulsar vs. non-pulsar)

- **Pipeline:** train/test split, 5-fold CV, upsample the positive class in training folds only (never validation/test).
- **GDA baseline** from scratch using the ϕ , μ_0 , μ_1 , Σ formulas.
- **Random Forest (scikit-learn)** tuned with RandomizedSearchCV (`n_estimators`, `max_depth`, `max_features`, `min_samples_split`).
- **Random Forest from scratch:** entropy, information gain, bootstrap sampling, random feature subsets. Keep depth and tree count small (the report could not tune this due to run time).
- **Evaluation:** accuracy, recall, precision, ROC and PR curves, and a threshold sweep (report moved from 0.5 to 0.2 to raise recall).
- *Optional:* SVM, or SMOTE vs. random oversampling.

Teammate 2: Unsupervised categorization and demo

- **Data:** train only on the 1,639 pulsar examples.
- **K-means baseline** ($k = 3$), with an elbow / silhouette check for choosing k .
- **Self-Organizing Map from scratch:** BMU search, neighborhood decay, learning-rate decay (NumPy or TensorFlow, 5x5 grid).
- **PCA for visualization only** (not as clustering input): cluster scatter plots and feature-to-component correlations.
- **Silhouette scores** for both clustering methods.
- **Demo:** notebook or small app (Streamlit/Gradio) where the user enters candidate features and gets a pulsar prediction plus a cluster assignment.

Deliverables split

Item	Teammate 1	Teammate 2
Code and repo	Supervised modules, requirements.txt	Unsupervised modules, demo
README	Setup and how to run the supervised part	How to run the unsupervised part and the demo
Write-up (PDF)	Problem statement, dataset, supervised approach and results	Unsupervised approach and results, conclusions and limitations
Slides	Intro, dataset, supervised methods and results	Unsupervised methods and results, demo, conclusions
Live demo	Runs and explains the supervised pipeline	Runs the demo app and the clustering

Timeline

Day	Work
Tue 29 - Wed 30 Sep	Repo setup, EDA, shared preprocessing; then start own modules.
Thu 1 - Fri 2 Oct	Core implementations finished (GDA and RF; K-means and SOM).
Sat 3 - Sun 4 Oct	Tuning, plots, metrics, demo app, first draft of write-up and slides.

Day	Work
Mon 5 Oct	Merge, clean repo, full rehearsal. Reviews run Oct 5-9, so be ready early.
Sat 10 Oct, 11:59 PM	Final deadline: repo link and PDF write-up.

Tips

- Marks depend on individual contribution and Q&A: commit your own code regularly, and spend an hour walking each other through your half so both can answer questions on everything.
- The brief says "One-Page" in the heading but "two-page" in the description. Keep the write-up to two pages at most and confirm with faculty if unsure.
- Use the report's limitations as conclusions: clustering scored only 0.44 silhouette because integrated-profile features hide frequency information, and the from-scratch RF was too slow to tune.